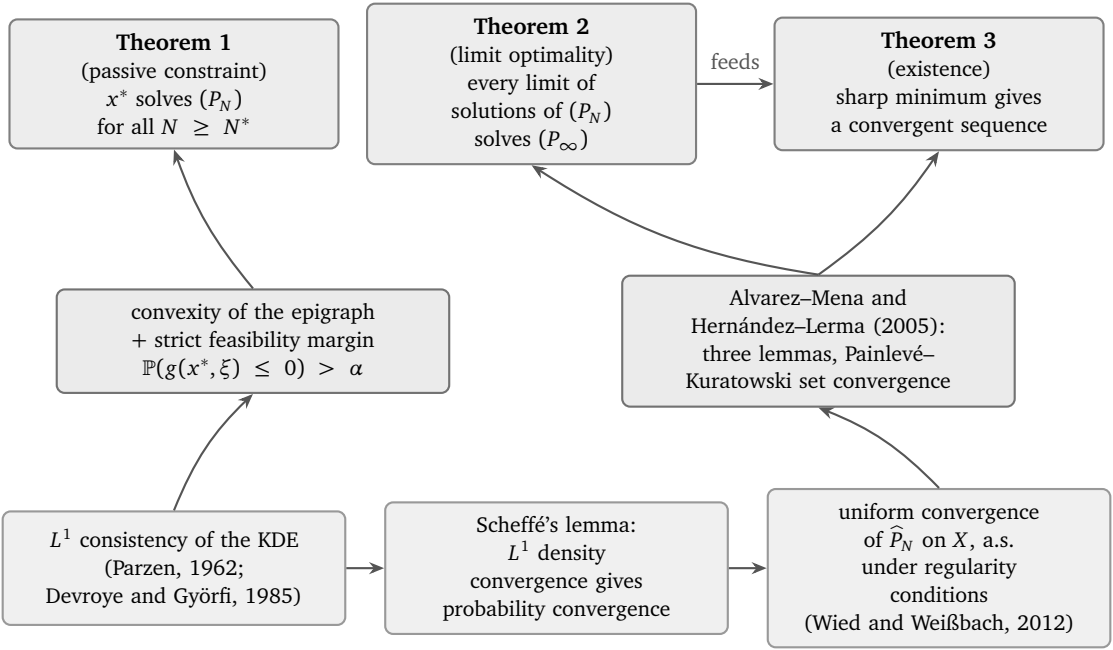
Theorem 1
(passive constraint)
x^* solves (P_N)
for all $N \geq N^*$
Theorem 2
(limit optimality)
every limit of
solutions of (P_N)
solves (P_∞)
feeds
Theorem 3
(existence)
sharp minimum gives
a convergent sequence
convexity of the epigraph
+ strict feasibility margin
$\mathbb{P}(g(x^*, \xi) \leq 0) > \alpha$
Alvarez–Mena and
Hernández–Lerma (2005):
three lemmas, Painlevé–
Kuratowski set convergence
L^1 consistency of the KDE
(Parzen, 1962;
Devroye and Györfi, 1985)
Scheffé's lemma:
L^1 density
convergence gives
probability convergence
uniform convergence
of $\widehat{P}_N$ on X, a.s.
under regularity
conditions
(Wied and Weißbach, 2012)